

Einstein Velocity Addition from Associativity and Boundary–Fixing Möbius Translations

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November 5, 2025

Abstract

We give an axiomatic characterization of the one–dimensional Einstein velocity addition law

$$u \oplus v = \frac{u + v}{1 + \frac{uv}{c^2}}$$

that does not presuppose Lorentz matrices or coordinates. On the bounded open interval $I_\kappa = (-\frac{1}{\kappa}, \frac{1}{\kappa})$ (with $\kappa = 1/c > 0$), assume: (i) a continuous, strictly increasing, associative composition with identity 0 and inverses; and (ii) each left translation extends to a real–analytic Möbius map of $\widehat{\mathbb{R}} = \mathbb{R} \cup \{\infty\}$ fixing the boundary points $\pm\frac{1}{\kappa}$. We prove that the unique (up to positive scaling) linearizer is

$$f(x) = \frac{1}{\kappa} \operatorname{artanh}(\kappa x),$$

hence $x \oplus y = f^{-1}(f(x) + f(y)) = (x + y)/(1 + \kappa^2 xy)$. This derivation clarifies exactly which structural properties uniquely determine relativistic velocity composition, independent of Lorentz matrix algebra.

MSC 2020: 39B72, 22E10, 83A05. *Keywords:* associative law, Aczél representation, Möbius maps, $\operatorname{PSL}(2, \mathbb{R})$, rapidity, Einstein addition.

1 Introduction

Let $\kappa > 0$ and $I_\kappa = (-\frac{1}{\kappa}, \frac{1}{\kappa})$. We consider a binary operation $\oplus : I_\kappa^2 \rightarrow I_\kappa$ whose left translations $\tau_a(x) := a \oplus x$ extend as real–analytic Möbius maps of $\widehat{\mathbb{R}}$ fixing the endpoints $\pm\frac{1}{\kappa}$. Our main result shows that these hypotheses uniquely determine the rapidity chart and hence the Einstein law.

Theorem 1.1 (Main classification). *Suppose $\oplus : I_\kappa^2 \rightarrow I_\kappa$ satisfies:*

- (A1) **Regular associative law:** *$\oplus$ is continuous, strictly increasing in each variable, associative, has identity 0, and each x has an inverse $-x$.*
- (A2) **Boundary–fixing Möbius translations:** *For each $a \in I_\kappa$, the map $\tau_a(x) := a \oplus x$ extends to a real–analytic fractional linear map of $\widehat{\mathbb{R}}$ with $\tau_a(\pm\frac{1}{\kappa}) = \pm\frac{1}{\kappa}$.*

Then there exists a strictly increasing $f : I_\kappa \rightarrow \mathbb{R}$, unique up to multiplication by a positive constant, such that

$$x \oplus y = f^{-1}(f(x) + f(y)), \quad f(x) = \frac{1}{\kappa} \operatorname{artanh}(\kappa x).$$

Equivalently,

$$x \oplus y = \frac{x + y}{1 + \kappa^2 xy}, \quad x, y \in I_\kappa.$$

Remark on scope and minimality. By Aczél's theorem on continuous, strictly monotone associative laws on an interval, $\oplus$ is order-isomorphic to $(\mathbb{R}, +)$; commutativity need not be assumed separately. The content of (A2) is that the Möbius *boundary-fixing* condition uniquely selects the chart f (up to scale), namely the rapidity $f = (1/\kappa) \operatorname{artanh}(\kappa \cdot)$. *What the theorem asserts is precisely that hypothesis (A2) forces this linearizer.*

We furnish two short, complementary proofs. Proof I uses a global conjugacy (no differentiability of $a \mapsto \tau_a$ is required). Proof II identifies the unique quadratic generator for a boundary-fixing Möbius flow and integrates $f'(x) = 1/V(x)$.

2 Preliminaries: Aczél representation and Möbius basics

We recall two standard facts.

Lemma 2.1 (Aczél representation). *If $M : E^2 \rightarrow E$ is continuous, associative, and strictly increasing in each variable on a real interval E , then there exists a continuous strictly monotone $f : E \rightarrow \mathbb{R}$, unique up to positive affine change, such that $M(x, y) = f^{-1}(f(x) + f(y))$. If $0 \in E$ is a neutral element then $f(0) = 0$.*

Lemma 2.2 (Boundary-fixing Möbius maps). *Let $p < q$ and let T be an orientation-preserving Möbius transformation of $\widehat{\mathbb{R}}$ with $T(p) = p$, $T(q) = q$. Then the coordinate*

$$r(x) = \frac{1}{2} \log \left(\frac{x - p}{q - x} \right)$$

conjugates T to a translation: $r \circ T = r + \lambda$ for some $\lambda \in \mathbb{R}$ (depending on T).

Sketch. After an affine change sending (p, q) to $(-1, 1)$, T is conjugate in $\operatorname{PSL}(2, \mathbb{R})$ to a diagonal matrix; r is the log cross-ratio with $\{p, q\}$.

3 Proof I (Conjugacy/Cayley): boundary form forces rapidity

Normalize $\kappa = 1$ (restore by rescaling at the end). Set $I = (-1, 1)$, $p = -1$, $q = 1$. By Lemma 2.1 there exists a strictly increasing $f : I \rightarrow \mathbb{R}$ with $f(0) = 0$ such that

$$x \oplus y = f^{-1}(f(x) + f(y)), \quad \tau_a(x) = f^{-1}(f(x) + f(a)).$$

By assumption, each τ_a extends to a Möbius map fixing ± 1 . By Lemma 2.2 there is a coordinate $r(x) = \frac{1}{2} \log \frac{1+x}{1-x} = \operatorname{artanh}(x)$ for which

$$r(\tau_a(x)) = r(x) + \lambda(a),$$

with $\lambda(a)$ independent of x . Comparing with the Aczél form,

$$r(f^{-1}(f(x) + f(a))) = r(x) + \lambda(a).$$

Setting $x = 0$ and using $f(0) = 0$ gives $\lambda(a) = r(f^{-1}(f(a))) = r(a)$, so

$$r(f^{-1}(f(x) + t)) = r(x) + t \quad \text{for all } x \in I, t \in f(I).$$

Thus $r \circ f^{-1}$ conjugates translations to translations, hence is an affine map. Monotonicity and the normalization at 0 imply $r = \alpha f$ with $\alpha > 0$, i.e. $f = \alpha^{-1} \operatorname{artanh}$. Absorb the scale into f (Aczél representation is unique up to positive scaling) to get $f = \operatorname{artanh}$. Therefore $x \oplus y = \tanh(\operatorname{artanh} x + \operatorname{artanh} y) = (x + y)/(1 + xy)$. Restoring κ via $x \mapsto \kappa x$ yields the stated formula.

4 Proof II (Infinitesimal): the quadratic generator $V(x) = c_0(1 - \kappa^2 x^2)$

Again normalize $\kappa = 1$. Assume $a \mapsto \tau_a$ is C^1 at $a = 0$ (this mild regularity is not needed in Proof I). Then $\{\tau_a\}$ is a C^1 one-parameter subgroup of $\text{PSL}(2, \mathbb{R})$ fixing ± 1 . The Lie algebra of real Möbius vector fields on the line is spanned by ∂_x , $x\partial_x$, $x^2\partial_x$, so the infinitesimal generator has the quadratic form

$$V(x) = \alpha + \beta x + \gamma x^2.$$

The boundary conditions $V(\pm 1) = 0$ force $\beta = 0$ and $\alpha = -\gamma$, hence $V(x) = c_0(1 - x^2)$ with $c_0 > 0$ for an increasing flow. In the Aczél chart $u = f(x)$, translations are $u \mapsto u + t$. Pulling back and differentiating at $t = 0$ gives

$$V(x) = \left. \frac{d}{dt} \right|_{t=0} f^{-1}(f(x) + t) = (f^{-1})'(f(x)) = \frac{1}{f'(x)}.$$

Thus $f'(x) = \frac{1}{c_0(1-x^2)}$, so $f(x) = \frac{1}{c_0} \text{artanh}(x)$ with $f(0) = 0$. Reparametrize the subgroup (absorb c_0 into the parameter) to set $c_0 = 1$. Therefore $f = \text{artanh}$ and $\oplus$ is Einstein addition; rescale $x \mapsto \kappa x$ to restore general κ .

5 Consequences and normal forms

Rapidity/log-odds forms. Using $\text{artanh } z = \frac{1}{2} \log \frac{1+z}{1-z}$,

$$f(x) = \frac{1}{2\kappa} \log \left(\frac{1 + \kappa x}{1 - \kappa x} \right), \quad x \oplus y = \tanh(f(x) + f(y)) = \frac{x + y}{1 + \kappa^2 xy}.$$

Small-signal expansion. Expanding

$$\frac{x + y}{1 + \kappa^2 xy} = (x + y)(1 - \kappa^2 xy + \kappa^4 x^2 y^2 - \dots)$$

gives the cubic correction

$$x \oplus y = x + y - \kappa^2(x^2 y + x y^2) + O(\|(x, y)\|^5).$$

General bounded intervals. If an operation on (a, b) has left translations extending to Möbius maps fixing the endpoints $a < b$, an affine change sends (a, b) to I_κ and reduces to Theorem 1.1.

6 Context and remarks

Classification in $\text{PSL}(2, \mathbb{R})$. Elements of $\text{PSL}(2, \mathbb{R})$ are elliptic, parabolic, or hyperbolic (by trace). Hypothesis (A2) singles out the hyperbolic case with two real fixed points, i.e. translations along a geodesic in the disk/upper half-plane model. The rectifying coordinate is (up to scale) rapidity.

Historical note. Varičák emphasized the hyperbolic geometry of velocity space a century ago; rapidity has appeared repeatedly since as the linearizing parameter for 1D boosts. Modern expositions of Einstein addition via Ungar's gyrogroup formalism place this in a broader noncommutative setting in higher dimensions.

No need to assume commutativity. Because Aczél puts $(I_\kappa, \oplus)$ in order-isomorphism with $(\mathbb{R}, +)$ under (A1), commutativity is automatic.

On regularity. Proof I uses only global conjugacy of boundary-fixing Möbius maps to translations; the mild C^1 assumption appears only in Proof II to speak of generators.

7 Open question (higher dimensions)

In 1D the law is forced by (A1)–(A2). In higher dimensions, velocity composition is noncommutative/gyrocommutative; Ungar’s gyrogroups capture this structure. A natural question is to formulate minimal axioms on a bounded domain in $\mathbb{R}^n$ whose left translations extend to (boundary-fixing) conformal automorphisms of the ball, recovering Einstein addition as the unique normal form.

Acknowledgments. I thank colleagues and online discussants for pointers to classical sources on Möbius conjugacy and rapidity.

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